

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Mathematics

Questions: 49

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

0 - 1 1 . Hence find A-1. I 1 0 (b) Show that $\int_0^{\pi/2} \sin^2 x \cos^2 x dx = \frac{\pi}{16}$

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2. [UNKNOWN] paper-i

00 . 2 , frc coordinates. Hence show that $\int_0^{\pi/2} e^{-x} dx = \frac{1}{2}$

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3. [UNKNOWN] paper-i

0. 10 $e^{3x}(dy)^{\pm} dx$ A planet is describing an ellipse about the Sun as a focus. Show that its velocity away from the Sun is the greatest when the radius vector to the planet is at a right angle to the major axis of path and that the velocity a . 2ic e then is , where 2a is the major axis, e is the eccentricity and T TV1— e² is the periodic time. 10 A semi-ellipse bounded by its minor axis is just immersed in a liquid, the density of which varies as the depth. If the minor axis lies on the surface, then find the eccentricity in order that the focus may be the centre of pressure. Evaluate if A where S is the surface of the cone, $z = 2 - \sqrt{x^2 + y^2}$ above xy-plane and A A A f = (x — z) i + (x³ + yz) j — 3xy² k . 10 10 FSI-P-MTH 6
===== PAGE 7 ===== d2y

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4. [UNKNOWN] paper-i

Q1. (a) Let A be a square matrix of order 3 such that each of its diagonal elements is 'a' and each of its off-diagonal elements is 1. If B = bA is orthogonal, determine the values of a and b. 8 (b) Let V be the vector space of all 2 x 2 matrices over the field R. Show that W is not a subspace of V, where (i) W contains all 2 x 2 matrices with zero determinant. W consists of all 2 x 2 matrices A such that A² = A. 8 (c) Using the Mean Value Theorem, show that f(x) is constant in [a, b], if f'(x) = 0 in [a, b]. f(x) is a decreasing function in (a, b), if f'(x) exists and is < 0 everywhere in (a, b). (d) Let u(x, y) = ax² + 2hxy + by² and v(x, y) = Ax² + 2Hxy + By². Find the Jacobian J = acu' v) , and hence show that u, v are independent unless a(x, y) a b h ABH = (e) Find the equations of the planes parallel to the plane 3x - 2y + 6z + 8 = 0 and at a distance 2 from it. 8

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5. [UNKNOWN] paper-i

1. then prove that the total angle through which , 2 the cube can swing without falling is 90°. 8 Prove that $v^{2rn} = n(n+1)$ ril-2 -> -> n and that rr is irrotational, where $r = \sqrt{x^2 + y^2 + z^2}$. 8

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6. [UNKNOWN] paper-i

Q2. (a) State the Cayley-Hamilton theorem. Verify this theorem for the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

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7. [UNKNOWN] paper-i

2) 8 8 FSI-P-MTH 2 ===== PAGE 3 ===== Hence evaluate the following integrals: $\int_0^{\pi/2} \sin^4 x \cos^5 x dx$ $\int_0^1 x^3(1-x^2)^{5/2} dx$ 1 x4(1— x)3 dx 10 Find the maxima and minima for the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$. Also find the saddle points (if any) for the function. 10 Show that the angles between the planes given by the equation $2x^2 - y^2 + 3z^2 - xy + 7zx + 2yz = 0$ is $\tan^{-1} \frac{1}{\sqrt{2}}$

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8. [UNKNOWN] paper-i

2. Find the angle between the lines whose direction cosines are given by the relations $l + m + n = 0$ and $2/m + 2/n - mn = 0$. /0 (i) 0 1 FSI-P-MTH 3 ===== PAGE 4 =====

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9. [UNKNOWN] paper-i

2. -8x D = + - y = e where $8 \frac{dx}{dx} \cdot \text{equation } d^2Y \ 2x \ 4y = x^2 - x^4$. 8 cbc2 dx A particle is undergoing simple harmonic motion of period T about a centre O and it passes through the position P (OP = b) with velocity v in the direction OP. Prove that the time that elapses before it returns to P . T - is-tan vir 8 it K2mb A heavy uniform cube balances on the highest point of a sphere whose radius is r. If the sphere is rough enough to prevent sliding and if the side of the cube be 71±

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10. [UNKNOWN] paper-i

Q3. (a) Reduce the following matrix to a row-reduced echelon form and hence find its rank : $\begin{bmatrix} 10 & -1 & 2 & -1 & 2 & 4 & 2 \\ 5 & 1 & 6 & 3 & 2 & 2 & 2 \end{bmatrix}$ A = $\begin{bmatrix} 0 & 0 & 1 \\ 5 & 1 & 6 & 3 & 2 & 2 & 2 \end{bmatrix}$ Given that the set $\{u, v, w\}$ is linearly independent, examine the sets (i) $\{u + v, v + w, w + u\}$ (ii) $\{u + v, u - v, u - 2v + 2w\}$ for linear independence. /0 Evaluate the integral $\int e^{-(x^2 + y^2)} dx dy$

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11. [UNKNOWN] paper-i

Q4. (a) Find A = a matrix A function the eigenvalues $\lambda_1 = 0, \lambda_2 = -2, \lambda_3 = 1, \lambda_4 = 3$ D (if it f(x, and the corresponding eigenvectors for the matrix . Examine whether the matrix A is diagonalizable. Obtain is diagonalizable) such that $D = P^{-1} A P$. /0 y) is defined as follows: $x^2 + y^2 = 0$ if $(x, y) = (0, 0)$ Show that $f_{xy}(0, 0) = f_{yx}(0, 0)$. /0 Find the equation of the right circular cone with vertex at the origin and whose axis makes equal angles with the coordinate axes and the generator is the line passing through the origin with direction ratios (1, -2, 2). /0 Find the shortest distance between distance and the equation of the line of the shortest $x - 3y - 8z = 3$ the lines $x + 3y + 7z = 6$ /0 -3 2 4 FSI-P-MTH 4 ===== PAGE 5 ===== SECTION B

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12. [UNKNOWN] paper-i

Q5. Solve (2D3 - 7D2) Solve the differential 7D

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13. [UNKNOWN] paper-i

Q6. Solve the differential equation $(-dy^2 + 2x \frac{dy}{dx} - y^2) \cot x = y^2$. 15 dxJ dx A string of length a, forms the shorter diagonal of a rhombus formed of four uniform rods, each of length b and weight W, which are hinged together. If one of the rods is supported in a horizontal position, then prove that the tension 2W (2b2 -a2) i of the string s 10 b-j4b2 -a2 FSI-P-MTH 5 ===== PAGE 6 ===== (c) Using Stokes' theorem, evaluate $\oint_C [(x + y) dx + (2x - z) dy + (y + z) dz]$, where C is the boundary of the triangle with vertices at (2, 0, 0), (0, 3, 0) and (0, 0, 6). 15

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14. [UNKNOWN] paper-i

Q7. (a) Solve the differential equation $3(u \cdot y) e^{2y}$

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15. [UNKNOWN] paper-i

Q8. (a) Solve $y' + 4y = \tan 2x$ by using the method of variation of parameter. /0 dx2 A particle moves in a straight line, its acceleration directed towards a fixed point O in the line and is always equal to u^2/x , when it is at a distance x from O. If it starts from rest at a distance a from O, then prove that it will arrive at O with a velocity $\sqrt{2}u$, after time $\frac{2a}{u}$. 8 11-6 /0 15 p. Find the curvature and torsion of the circular helix $r = a(\cos \theta, \sin \theta, \theta \cot \alpha)$, is the constant angle at which it cuts its generators. /0 If the tangent to a curve makes a constant angle α , with a fixed line, then prove that $\cos \alpha \pm \sin \alpha = 0$. Conversely, if $\frac{1}{\cos \alpha}$ is constant, then show that the tangent makes a constant angle with a fixed direction. /0

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16. [UNKNOWN] paper-ii

0. $u = \frac{1}{2} \nabla^2 \psi$, $v = -\frac{1}{2} \frac{\partial \psi}{\partial y}$, $w = \frac{1}{2} \frac{\partial \psi}{\partial x}$ by the to Euler-Lagrange dimensional equations of motion. 12 velocity field: 8 equations of vortex lines. 8 the conditions 12 1+t The velocity vector in the flow field is given $= (az - by)i + (bx - cz)j + (cy - ax)k$ where a, b, c are non-zero constants. Determine $a^2u + a^2v$ Solve Laplace's 0 equation $5x^2 + 8y^2 = 1$ subject $u(0, y) = u(1, y) = u(x, 0) = 0$ and $u(x, a) = \sin(\pi x/a)$.

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17. [UNKNOWN] paper-ii

1. (a) Prove that every group of order four is Abelian. 8

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18. [UNKNOWN] paper-ii

1. (b) A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as below: $f(x) = 1$ if x is rational $f(x) = 0$ if x is irrational Prove that f is continuous at $x = 0$ but discontinuous at all other points in $\mathbb{R}$. 10 1 fsi-d-mth ===== PAGE 2 =====

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19. [UNKNOWN] paper-ii

1. (c) If $f(z) = u(x, y) + iv(x, y)$ is an analytic function of $z = x + iy$ and $u + 2v = x^3 - 2y^3 + 3xy(2x - y)$ then find $f(z)$ in terms of z . 8

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20. [UNKNOWN] paper-ii

1. (d) Solve by simplex method the following LPP : Minimize $Z = x_1 - 3x_2 + 2x_3$ subject to the constraints $3x_1 - x_2 \leq 2x_3$ 'S. $7 \leq 2x_1 + 4x_2 \leq 12$ $4x_1 + 3x_2 + 8x_3 \leq 10$ and $x_1, x_2, x_3 \geq 0$

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21. [UNKNOWN] paper-ii

1. (a) Prove that a non-commutative group of order $2n$, where n is an odd prime, must have a subgroup of order n . 8 A function $f: [0, 1] \rightarrow [0, 1]$ is continuous on $[0, 1]$. Prove that there exists a point c in $[0, 1]$ such that $f(c) = c$. 10 If $u = (x - 1)^3 - 3xy^2 + 3y^2$, determine v so that $u + iv$ is a regular function of $x + iy$. 10 Solve by simplex method the following Linear Programming Problem : 12 Maximize $Z = 3x_1 + 2x_2 + 5x_3$ subject to the constraints $2x_1 + x_2 + 5x_3 \leq 430$ $3x_1 + 2x_2 + 5x_3 \leq 460$ $x_1 + 4x_2 + 5x_3 \leq 420$ $x_1, x_2, x_3 \geq 0$

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22. [UNKNOWN] paper-ii

2. (a) Let G be the set of all real numbers except -1 and define $a * b = a + b + ab$ $\forall a, b \in G$. Examine if G is an Abelian group under $*$. 10

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23. [UNKNOWN] paper-ii

2. (b) Let H and K are two finite normal subgroups of co-prime order of a group G . Prove that $hk = kh$ $\forall h \in H$ and $k \in K$. 10

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24. [UNKNOWN] paper-ii

2. (c) Let A be an ideal of a commutative ring R and $B = \{x \in R : x^n \in A \text{ for some positive integer } n\}$. Is B an ideal of R ? Justify your answer. 10

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25. [UNKNOWN] paper-ii

2. (d) Prove that the ring $Z[i] = \{a + ib : a, b \in \mathbb{Z}, i = \sqrt{-1}\}$ of Gaussian integers is a Euclidean domain. 10

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26. [UNKNOWN] paper-ii

2. (a) Find all the homomorphisms from the group $(\mathbb{Z}, +)$ to $(\mathbb{Z}_4, +)$. 10 Consider the function f defined by $f(x, y) = x^2 - y^2 + 2xy$ where $x^2 + y^2 \neq 0$. Show that $f(x, y) \neq f(y, x)$ at $(0, 0)$. 10 Prove that $\cos^2 x dx = 0$ $\sin x dx = 0$. 10 Let R be a commutative ring with unity. Prove that an ideal P of R is prime if and only if the quotient ring R/P is an integral domain. 10

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27. [UNKNOWN] paper-ii

3. (a) Evaluate $\int_0^1 (1-x)^2 dx$ and $f(x, 0)$ given that $x^2 \tan^{-1} x - y^2 \tan^{-1} y = x$ if $xy \neq 0$, $f(x, 0) = x$, otherwise 0. 3. (b) Find the maximum and minimum values of $x^2 + y^2 + z^2$ subject to the condition $x^2 y^2 z^2 = 10^{-4}$. 5 25

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28. [UNKNOWN] paper-ii

3. (c) Prove that $\sum_{n=0}^{\infty} x^n$ is convergent but not absolutely convergent. 12 (d) Find the volume of the region common to the cylinders $x^2 + y^2 = a^2$ and $xz = az$. 8 (a) Prove by the method of contour integration that $\int_0^{2\pi} \cos^2 \theta d\theta = \pi$. 12 $5 + 4\cos\theta$

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29. [UNKNOWN] paper-ii

3. (a) Find the minimum value of $x^2 + 4y^2 + 2z^2$ subject to the condition $ax + by + cz = 10$. Show by an example that in a finite commutative ring, every maximal ideal need not be prime. 10 (b) Evaluate the integral $\int_0^{\pi} \cos^2 x dx$, where n is a positive integer.

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30. [UNKNOWN] paper-ii

4. (b) Find the sum of residues of $f(z) = \frac{1}{\sin z}$ at its poles inside the circle $|z| = 2$. 8 $\cos z$ fsi-d-mth 2 ===== PAGE 3 =====
• co Evaluate $\int_0^{\pi} x \sin x dx$ A computer programs to x r x $y=0$ be be the centre r y c d 8 has four expert programmers. The centre needs four application developed. The head of the centre after studying carefully the developed, estimates the computer times in hours required by the application programs as follows: programs to experts to Programs ABCD 5 3 2 8 7 9 2 6 6 4 5 7 5 7 7 8 6. (a)

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31. [UNKNOWN] paper-ii

4. (a) Show that $\frac{dy}{dx} = \frac{f(x)}{g(x)}$ where $f(x) = \frac{1}{x}$, $g(x) = x^2$ taken over R : the triangle bounded by $x = 0$, $y = 0$, $x + y = 1$. 10 Let $f_n(x) = x^n$ $E[0, 1]$. Show that the sequence $\{f_n\}$ is uniformly $n + x^2$ convergent on $[0, 1]$. 8 Let H be a cyclic subgroup of a group G . If H be a normal subgroup of G , prove that every subgroup of H is a normal subgroup of G . 10 The capacities of three production facilities S_1 , S_2 and S_3 and the requirements of four destinations D_1 , D_2 , D_3 and D_4 and transportation costs in rupees are given in the following table:

	D_1	D_2	D_3	D_4	Capacity
S_1	19	30	50	10	7
S_2	70	30	40	60	9
S_3	40	8	70	20	18
Demand	5	8	7	14	34

 Find the minimum transportation cost using Vogel's Approximation Method (VAM). 12 SECTION—B

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32. [UNKNOWN] paper-ii

5. (a) Find the partial differential equation of all planes which are at a constant distance a from the origin. 10 (b) A solid of revolution is formed by rotating about the x -axis, the area between the x -axis, the line $x = 0$ and a curve through the points with the following coordinates:

x	0.0	0.25	0.50	0.75	1.00
F_2	1.50	1.0	0.9896	0.9589	0.9089
V	0.8415	0.8029	0.7635		

 Estimate the volume of the solid formed using Weddle's rule. 10 Write a program in BASIC to multiply two matrices (checking for consistency for multiplication is required). 10 (d) Air, obeying Boyle's law, is in motion in a uniform

tube of small section. Prove that if p be the density and v be the velocity at a distance x from a fixed point at a^2n a^2 time t , then $\frac{d}{dt} \{p(v^2 + k)\} = 0$. at2 ax2 10 FSI-D-MTH/47 3 [P.T.O. ===== PAGE 4 =====

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33. [UNKNOWN] paper-ii

6. (b) Assign the programs to the programmers in such a way that total computer time is least. 12 SECTION '13' Form the partial differential equation by eliminating arbitrary functions y_0 and from the relation $z = (x^2 - y)^2 + (x^2 + y)$. 8 Write a BASIC program to compute the multiplicative inverse of a non-singular square matrix. 12 A uniform rectangular parallelopiped of mass M has edges of lengths $2a$, $2b$, $2c$. Find the moment of inertia of this rectangular parallelopiped about the line through its centre parallel to the edge of length $2a$. 10 Evaluate $\int_0^1 \frac{dx}{x^2 + 1}$ using the composite trapezoidal rule with four decimal precision, i.e., with the absolute value of the error not exceeding 5×10^{-5} . 10 Solve the partial differential equation: $az^2 \frac{\partial z}{\partial x} + (x - y) \frac{\partial z}{\partial y} = 2xz$ ax ay Find the surface which is orthogonal to the family of surfaces $z(x + y) = c(3z + 1)$ and which passes through the circle $x^2 + y^2 = 1$, $z = 1$. 8 az Find complete integral of $xp - yq = xq f(z - px - qy)$ where $p = -az$, $q = -$. 12 ax ay A tightly stretched string with fixed end points $x = 0$ and $x = 1$ is initially in a position given by $y = y_0 \sin 3x$. It is released from rest from this position, find / the displacement $y(x, t)$. 12 3 fsi-d-mth

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34. [UNKNOWN] paper-ii

6. (a) Find the complete integral of the partial differential equation $(p^2 + q^2)x = zp$ and deduce the solution which passes through the curve $x = 0$, $z^2 = 4y$. Here $p = -$, $q = -$, ax ay az az 12 Apply fourth-order Runge-Kutta method to compute y at $x = 0.1$ and $x = 0.2$, given that $\frac{dy}{dx} = x + y^2$, $y = 1$ at $x = 0$. 12 dx For a particle having charge q and moving in an electromagnetic field, the potential energy is $U = q(4 - D \cdot A)$, where (1) and A are, respectively, known as the scalar and vector potentials. Derive expression for Hamiltonian for the particle in the electromagnetic field. 8 Write a program in BASIC to implement trapezoidal rule to compute $\int_0^1 e^{-x^2} dx$ with 10 subdivisions. 8

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35. [UNKNOWN] paper-ii

7. (a) Find the real root of the equation $x^3 + x^2 + 3x + 4 = 0$ correct up to five places of decimal using Newton-Raphson method. 10

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36. [UNKNOWN] paper-ii

7. (b) A river is 80 metre wide, the depth y , in metre, of the river at a distance x from one bank is given by the following table:

x	0	10	20	30	40	50	60	70	80
y	0	4	7	9	12	15	14	8	3

 Find the area of cross-section of the river using Simpson's rd rule. 10

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37. [UNKNOWN] paper-ii

7. (c) Find y for $x = 0.2$ taking $h = 0.1$ by modified Euler's method and compute the error, given that: $\frac{dy}{dx} = x + y$, $y(0) = 1$. 10 dx (d) (a) Assuming a 32 bit computer representation of signed integers using 2's complement representation, add the two numbers -1 and -1024 and give the answer in 2's complement representation. 10 Consider a mass m on the end of a spring of natural length 1 and spring constant k . Let y be the vertical coordinate of the mass as measured from the top of the spring. Assume that the mass can only move up and down in the Vertical direction. Show that 8.(b)

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38. [UNKNOWN] paper-ii

7. (a) Solve $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$, where $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$. ax ay If the solution of the above equation represents a sphere, what will be the coordinates of its centre? 8 (b) The velocity v (km/mm) of a moped is given at fixed interval of time (min) as below :

t	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1
v	1.758819	1.904497	2.049009	2.18874	2.31977	2.44987	2.57979	2.70985	2.83985	2.96985	3.09985

 Estimate the distance covered during the time (use Simpson's one-third rule). 10 Assuming a 16-bit computer representation of signed integers, represent -44 in 2's complement representation. 10 In the case of two-dimensional motion of a liquid streaming past a fixed circular disc, the

velocity at infinity is in a fixed direction, where u is a variable. Show that the maximum value of the velocity at any point of the fluid is $2u$. Prove that the force necessary to hold the disc is $2\pi\mu u$, where m is the mass of the liquid displaced by the disc. 12 a^2v a^2v

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39. [UNKNOWN] paper-ii

8. (d) = $\frac{m}{2} \frac{d^2x}{dt^2}$ Also determine and solve the corresponding Find the streamlines and pathlines of the two

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40. [UNKNOWN] paper-ii

8. (a) Find a real function V of x and y , satisfying $\nabla^2 V = -4n(x^2 + y^2)$ and ∇V reducing to zero, when $y = 0$. 10
FSI-D-MTH/47 4 ===== PAGE 5 ===== The equation $x^6 - x^4 - x^3 - 1 = 0$ has one real root between 1.4 and 1.5. Find the root to four places of decimal by regula-falsi method. 10 A particle of mass m is constrained to move on the inner surface of a cone of semi-angle α under the action of gravity. Write the equation of constraint and mention the generalized coordinates. Write down the equation of motion. 10 Two sources, each of strength m , are placed at the points $(-a, 0)$, $(a, 0)$ and a sink of strength $2m$ at the origin. Show that the streamlines are the curves $(x^2 + y^2)^2 = a^2(x^2 - y^2 + Axy)$ where A , is a variable parameter. Show also that the fluid speed at any point is $(2ma^2)/(r_1 r_2 r_3)$, where r_1, r_2, r_3 are the distances of the point from the sources and the sink. 10 * * * FSI-D-MTH/47 5 BS9-162

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41. [UNKNOWN] paper-ii

10 . 1 $\sin^{-1} x$ Show that the improper integral $\int_1^\infty \frac{1}{x^2} dx$ is convergent. 10 0 0 , where $x^2 + y^2 = 0$ FSI-D-MTH / 47 2 ===== PAGE 3 =====

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42. [UNKNOWN] unknown

1. 0 3-- $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \end{pmatrix}$ are congruent. 10 -1 1 3 _3 2 0 If l and w be two functions derivable in $[a, b]$ and $l'(x) = w(x)$ iff $w'(x) = l(x)$ for any x in this interval, then show that between two consecutive roots of $l(x) = 0$ in $[a, b]$, there lies exactly one root of $w(x) = 0$. 10 Show that the vectors $a_1 = (1, 0, -1)$, $a_2 = (1, 2, 1)$, $a_3 = (0, -3, 2)$ form a basis for R^3 . Express each of the standard basis vectors as a linear combination of a_1, a_2, a_3 . 10 FSI-P-MTH 2 ===== PAGE 3 =====

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43. [UNKNOWN] unknown

Q2. Find the equation of the cylinder whose generators are parallel to the line $-x = -z$ and whose guiding curve is $x^2 + y^2 = 4, z = 2$. 10 Show that the matrices $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

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44. [UNKNOWN] unknown

Q3. (a) Find the equation of the tangent plane that can be drawn to the sphere $x^2 + y^2 + z^2 - 2x + 6y + 2z + 8 = 0$, through the straight line $3x - 4y - 8 = 0 = y - 3z + 2$. 10 (b) If $f = f(u, v)$, where $u = ex \cos y$ and $v = ex \sin y$, show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2}$ 10 Let $T : V_2(R) \rightarrow V_2(R)$ be a linear transformation defined by $T(a, b) = (a, a + b)$. Find the matrix of T , taking $\{e_1, e_2\}$ as a basis for the domain and $\{(1, 1), (1, -1)\}$ as a basis for the range. 10 Evaluate $\iint_R (x^2 + xy) dx dy$ over the region R bounded by $xy = 1, y = 0, y = x$ and $x = 2$. 10

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45. [UNKNOWN] unknown

Q4. (a) Find the equations of the straight lines in which the plane $2x + y - z = 0$ cuts the cone $4x^2 - y^2 + 3z^2 = 0$. Find the angle between the two straight lines. 10 Show that the functions $u = x + y + z, v = xy + yz + zx$ and $w = x^3 + y^3 + z^3 - 3xyz$ are dependent and find the relation between them. 10 Find the locus of the point of intersection of the perpendicular $x^2 + y^2$ generators of the hyperbolic paraboloid $z = 2xy$. 10 (b) If $(n + 1)$ vectors $a_1, a_2, \dots, a_n, a$ form a linearly dependent set, then show that the vector a is a linear combination of $a_1, a_2, \dots, a_n$; provided $a_1, a_2, \dots, a_n$ form a

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46. [UNKNOWN] unknown

Q5. Find the complementary function $d^2y/dx^2 - y \sec x \cos x$ and particular integral for the equation $d^2y/dx^2 = 2 + \log x - y \sec x \cos x$ ($x > 0$) by the method of variation of d^2y/dx^2 and hence the general solution of $d^2y/dx^2 = 2 + \log x - y \sec x \cos x$.
 parameters. If the velocities in a simple harmonic motion at distances a , b and c from a fixed point on the straight line which is not the centre of force, are u , v and w respectively, show that the periodic time T is given by $u^2/v^2 = w^2/4$ (b) (c) $(a/b) \cdot (b/c) = 8/0$ c) $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$ From a semi-circle whose diameter is in the surface of a liquid, a circle is cut out, whose diameter is the vertical radius of the semi-circle. Find the depth of the centre of pressure of the remainder part. 8 $\rightarrow$ A A A If $r = xi + yj + zk$ and $f(r)$ is differentiable, show that $\text{div}[f(r)r] = rf(r) + 3f(r)$. $\rightarrow$ Hence or otherwise show that $\text{div}(-r/r^3) = 0$. 8

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47. [UNKNOWN] unknown

Q6. Solve the differential equation $(y^2 + 2xy) dx + (2x^2 - xy) dy = 0$. 10 Let T_1 and T_2 be the periods of vertical oscillations of two different weights suspended by an elastic string, and C_1 and C_2 are the statical extensions due to these weights and g is the acceleration due to gravity. $4m^2(C_1 - C_2)$ Show that $15gT_1^2 \cdot T_2^2 = 1/2 \rightarrow$ A Show that $F = (2xy + z^3)i + x^2j + 3xz^2k$ is a conservative force. Hence, find the scalar potential. Also find the work done in moving a particle of unit mass in the force field from $(1, -2, 1)$ to $(3, 1, 4)$. 15 FSI-P-MTH 4 ===== PAGE 5 =====

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48. [UNKNOWN] unknown

Q7. (a) The end links of a uniform chain slide along a fixed rough horizontal rod. Prove that the ratio of the maximum span to the length of the chain is $\mu \log 1 + (1+12)^2$ where μ is the coefficient of friction. 10 Solve: $10 dy = 4x + 6y + 5 dx$ $3y + 2x + 4$ A frame ABC consists of three light rods, of which AB, AC are each of length a , BC of length a , freely jointed together. It rests with BC horizontal, A below BC and the rods AB, AC over two smooth pegs E and F, in the same horizontal line, at a distance $2b$ apart. A weight W is suspended from A. Find the thrust in the rod BC. 10 Let a be a unit-speed curve in R^3 with constant curvature and zero torsion. Show that a is (part of) a circle. 10

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49. [UNKNOWN] unknown

Q8. (a) A solid hemisphere floating in a liquid is completely immersed with a point of the rim joined to a fixed point by means of a string. Find the inclination of the base to the vertical and tension of the string. A snowball of radius $r(t)$ melts at a uniform rate. If half of the mass of the snowball melts in one hour, how much time will it take for the entire mass of the snowball to melt, correct to two decimal places? Conditions remain unchanged for the entire process. For a curve lying on a sphere of radius a and such that the torsion is never 0, show that $\sqrt{2 + Kr^2} = a^2$. 15 15 10 FSI-P-MTH 5

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